Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Agentic AI · RAG Pipelines · LLM Evaluation · Distributed Systems · AWS · Full-Stack · CI/CD · Production Reliability · Mentoring

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years delivering production Python systems at enterprise scale - and two shipped agentic AI systems that demonstrate end-to-end AI product ownership. At Entrata, led delivery of 20+ production modules across 120+ Agile sprints: Python and JavaScript backends, event-driven distributed architecture (RabbitMQ), cloud-native AWS infrastructure, and CI/CD-driven quality at 99.5% uptime for 100K+ monthly active users. Owned HUD income-limit data pipeline end-to-end for 7 years - 180,000 domain records powering housing eligibility decisions for 100K+ units with 100% PHI compliance and zero data integrity incidents. Conducted 300+ code reviews and mentored 5+ engineers. AgentProbe is an open-source LLM evaluation framework: 5-metric quality engine, LLM-as-judge scoring, CI-integrated regression gate, multi-provider Anthropic/OpenAI/Ollama routing - zero regressions shipped across 1,240+ test cases. LeaseI is an 8-agent LangGraph stateful system with multi-source RAG retrieval, OCR document extraction, deterministic guardrails, HITL escalation, and source-linked answers - 93.6% precision, 87.9% F1 across 1,240 compliance test cases. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Engineering: Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, LLM Evaluation Frameworks, LLM-as-Judge, CI-Integrated Regression Gating, HITL, Guardrails, Structured Outputs, Prompt Engineering, ChromaDB, Vector Search, Embeddings, Anthropic API (Claude), OpenAI API, Ollama, Claude Code, MCP, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

Engineering & Infrastructure: Python (8yr), JavaScript, TypeScript, SQL, PHP, Bash, FastAPI, Flask, REST APIs, Docker, Microservices, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, NoSQL, PySpark, Linux, Git

Reliability & Data: Observability, Production Monitoring, Performance Engineering, Data Pipelines, ETL, Data Modeling, PHI Compliance (Libsodium), Feature Engineering, NumPy, Pandas, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) - multi-agent orchestration with RAG retrieval over distributed data sources, deterministic guardrails, HITL escalation, and CI/CD-integrated eval gate enforcing quality at every release.
- Leveraged Claude Code and AI-assisted development tools throughout the engineering lifecycle; built observability on agent performance, retrieval latency, and system reliability.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, and software engineering; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production systems and data pipelines at enterprise scale** - 20+ scalable Python and JavaScript modules at 99.5% uptime across 100K+ monthly active users; owned HUD income-limit pipeline 7 years (federal source → 180,000 domain records); RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS, Glue at enterprise scale.
- **AI-driven observability and secure system design** - AI-powered monitoring tooling reducing triage time 40%; ML-powered data validation across compliance-critical workflows; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); Redis caching; 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain).
- **Engineering leadership and quality** - 300+ code reviews enforcing engineering standards and documentation-based practices; maintained sub-10 bug count for 6+ consecutive months across multi-tenant production environments; CI/CD via GitHub Actions; mentored 5+ engineers on Python, distributed systems, and AI best practices; collaborated with product, compliance, and operations stakeholders to deliver measurable technical outcomes.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source LLM Evaluation Framework

Evals · CI/CD · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, multi-provider Anthropic/OpenAI/Ollama routing for cost optimization. Zero regressions shipped across 1,240+ test cases - evaluation infrastructure built before any prompt reaches production.

Leasel: 8-Agent Agentic Workflow Platform with RAG

Agentic · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system: 8-agent LangGraph stateful workflow with multi-step routing, planning, tool use, and state management; OCR document extraction and multi-source RAG retrieval; deterministic guardrails on every regulated output; HITL escalation on low-confidence decisions; source-linked answers with 100% citation correctness. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AI on FHIR: NLP-Driven Medical Records Query System

RAG · NLP · Full-Stack

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application (Python + React/TypeScript) translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017